

Sentiment and Emotion Datasets :: CHEAT SHEET

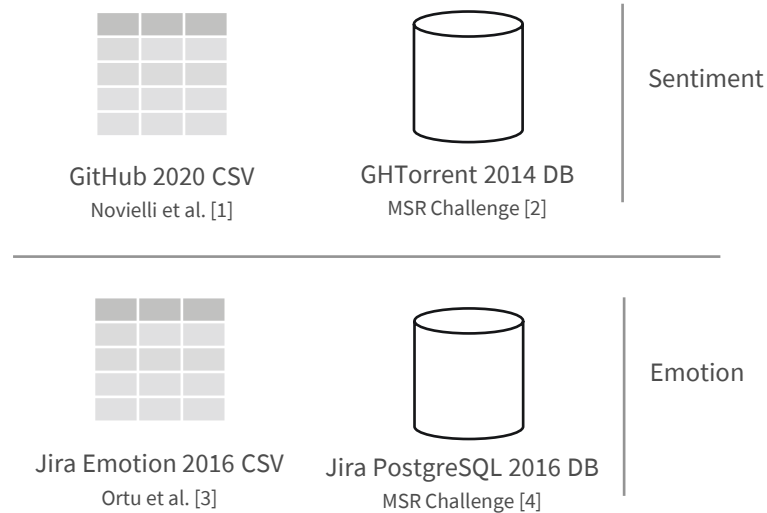


About

Manually annotated sentiment/emotion labels can be useful for training and evaluating novel machine learning classifiers. However, these labels lack sufficient metadata to be extended with new observations for the same projects that were created. This cheatsheet demonstrates the reconstruction process used to expand labels' metadata, define reproducible project configuration files, and make them usable in Kaiaulu for new analyses.

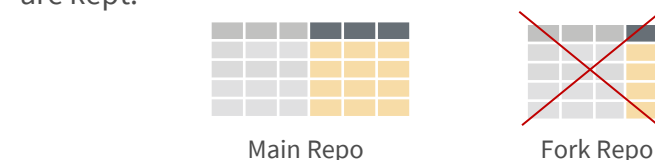
Raw Dataset

The following must be set up/downloaded:

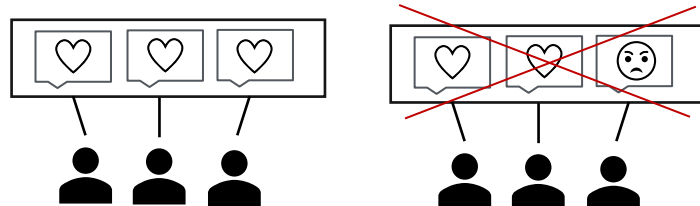


Label Selection Criteria

Sentiment labels span both main project repos and their forks. However, since forked repos contain a small fraction (< 10%) of labeled comments, only comments from the project's main repo are kept.

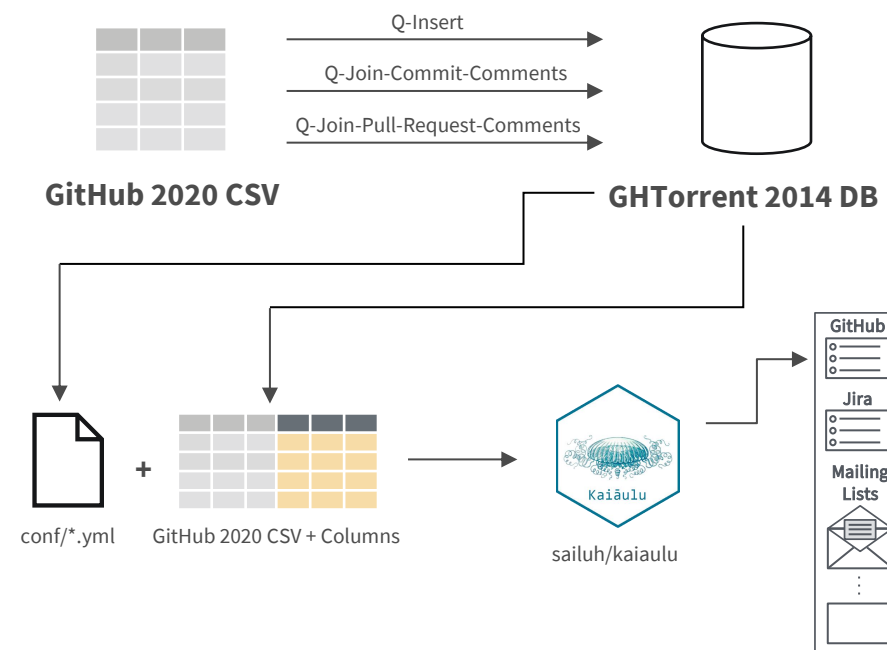


The Jira Emotion CSV contains emotion labels manually determined from multiple raters. Only comments with labels that show higher inter-rater agreement are kept.

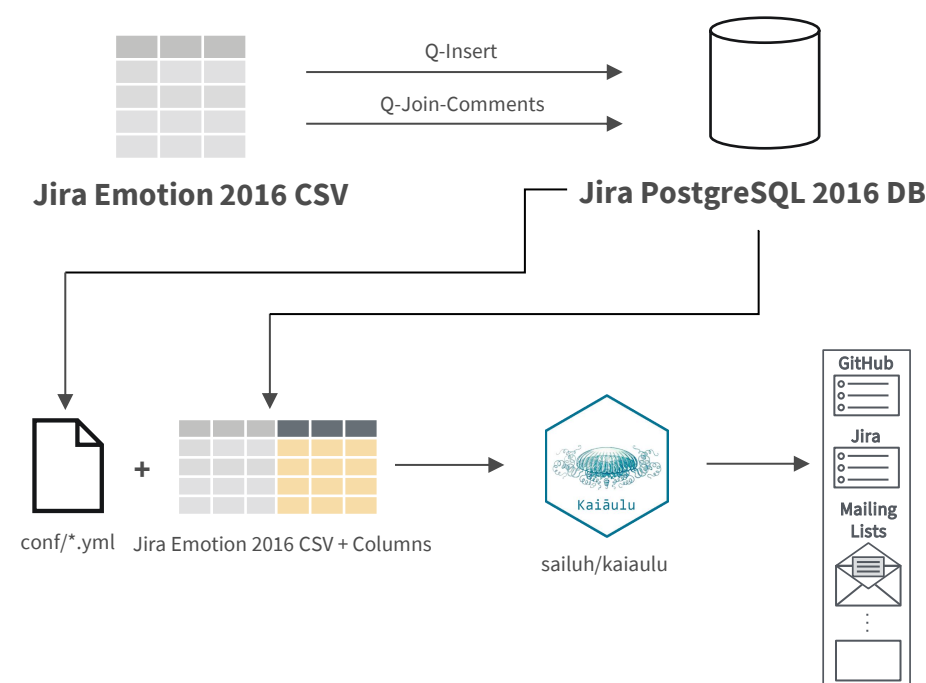


Reproducing Datasets with Kaiaulu

GitHub Sentiment Dataset



Jira Emotion Dataset



Related Notebooks

See supplementary notebooks that go in-depth on the documentation on making the Github dataset and Jira Emotion dataset compatible with Kaiaulu.

sailuh/sentiment_github_dataset

1. [1_load_sentiment_csv_to_mysql.ipynb](#)
2. [2_explore_relevant_projects.ipynb](#)
3. [3_scale_config_files.ipynb](#)
4. [4_add_sentiment_to_kaiaulu.ipynb](#)

sailuh/emotion_jira_dataset

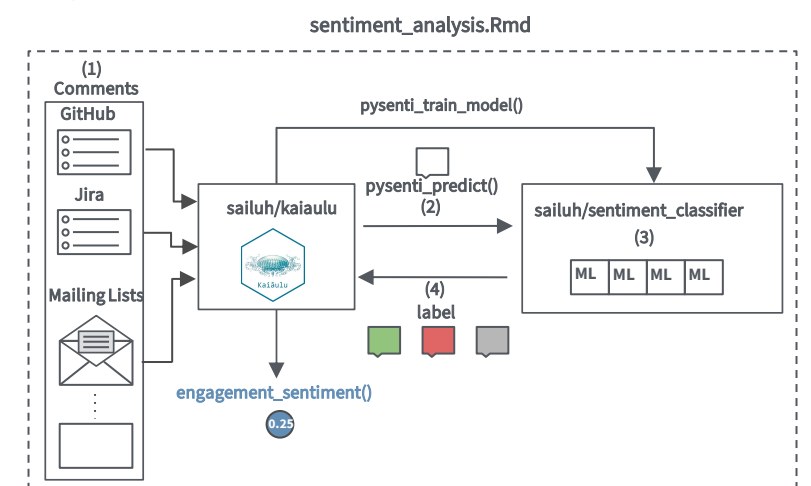
1. [1_emotion_csv_inter_rater_agreement.ipynb](#)

Data Expansion

Through this pipeline, we recover sufficient metadata to re-obtain the data using Kaiaulu. This metadata enables two types of dataset expansion using Kaiaulu:

1. **Temporal expansion:** Re-download the same projects' comments from 2014 through today.
2. **Horizontal expansion:** Download additional data sources for those same projects (e.g., source code, version control history) that can be linked back to sentiment labels.

The labeled comments can be used to train the sentiment classifier and for developer engagement (refer to sailuh/sentiment-classifier).



[1] N. Novielli, Fabio Calefato, Davide Dongiovanni, D. Girardi, and Filippo Lanubile, "Can We Use SE-specific Sentiment Analysis Tools in a Cross-Platform Setting?", Jun. 2020, doi: <https://doi.org/10.1145/3379597.3387446>.
[2] <https://web.archive.org/web/20210224080543/http://ansymore.uantwerpen.be/system/files/uploads/artefacts/alessandro/M-SR16/archive3.zip>
[3] M. Ortu et al., "The emotional side of software developers in JIRA," *Proceedings of the 13th International Conference on Mining Software Repositories*, May 2016, doi: <https://doi.org/10.1145/2901739.2903505>.
[4] <https://web.archive.org/web/20150206005357/http://ghtorrent.org/msr14.html>